



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2024-25 Summer Term Examination
Class: S. Y. B. Tech. (EXTC)

Date: 16/06/25
Time: 14:30 to 17:30

Subject: Digital circuits Design and
Microprocessors Interfacing
Subject Code: PCC-EC206

Max. Marks: 100
Duration: 3 hours

Note : 1. Use of calculator is not permitted. 2. Solve all questions

	Solve any 8 questions	Marks
Q.1	State difference between microprocessor and microcontroller. State advantages of microcontroller. Explain architecture of 8085 microprocessor.	10
Q.2	Design half and full adder. What do you mean by odd and even parity? Design 4 bit even parity generator using Gates.	10
Q.3	Draw and explain 4:1 MUX using Gates. Design Full subtractor using multiplexer. Implement basic gates using NAND and NOR gates?	10
Q.4	Convert decimal number 123 in to octal and hexadecimal numbers. Convert octal number 65.12 in to binary and decimal numbers. Construct 7-bit hamming code for 1101 data bits.	10
Q.5	Design 3 bit binary to gray code converter circuit. Draw logic diagram of twisted ring counter. State advantages of the same.	10
Q.6	Reduce the following functions using K map and implement using NAND Gates only. a. $F1 = \sum m(0,2,3,4,5,6,7,14,15)$ b. $F2 = \sum m(7,9,10,11,12,15) + d(0,1,2,3,4,5,6)$	10
Q.7	Explain propagation delay and Noise Margin. Design 3 bit synchronous counter.	10
Q.8	Explain different types of shift registers with logic diagrams and waveforms. State applications of shift registers. Explain different types of codes used in Digital Electronics.	10
Q.9	What do you mean by encoding and decoding. Design MOD 6 asynchronous counter using flip flops and draw waveforms.	10
Q.10	Define flip flop. Explain S-R, J-K, T and D flip flops. Design and explain 4 bit ripple counter using suitable waveforms	10
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Course Outcomes (COs):

CO1. fundamentals concepts used in digital circuits.

CO2. 8085 microprocessor architecture, digital codes and Boolean Algebra etc.

CO3. Design of adders, subtractors, code converter circuits such as binary to gray, Gray to binary, BCD to excess-3 and excess-3 to BCD etc.

CO4. Simplify logical functions using K map and Quine Mc Cluskey method, design combinational circuits

CO5. Design sequential circuits such as flip flops, counters and registers etc.

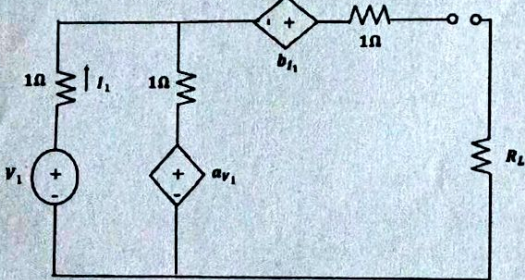
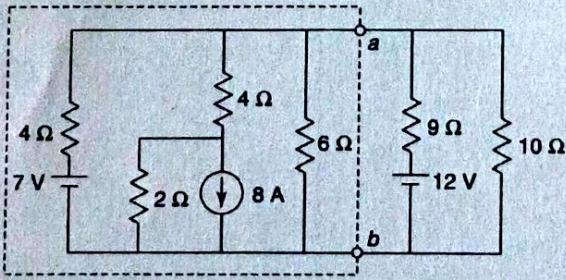
CO6. Memories such as RAM, ROM, EPROM etc

Bloom's Taxonomy: BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

		
Shri Guru Gobind Singhji Institute of Engineering and Technology, Vishnupuri, Nanded		
Department of Electronics and Telecommunication Engineering		
In-Semester Examination I Academic Year: 2025-26 (Semester -I)		
Class: SY B. Tech	Subject: Network Theory	Subject Code: PCC-EC203
Date: 21/08/2025	Time: 11:00 – 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Solve all questions.

Q. No.		Marks	CO	BT
1	What are different types of dependent sources?	2	CO1	BT1
2	Can two ideal voltage sources be connected in parallel and two ideal current sources be connected in series? Justify your answer.	2	CO1	BT2, BT3
3	For the following network , show that the equivalent Thevenin network is represented by $V_{\theta} = \frac{V_1}{2} (1 + a + b - ab) \text{ and } Z_{\theta} = \frac{3 - b}{2}$ 	7	CO2	BT4
OR				
	Find Norton's equivalent circuit for the portion of network shown in Fig. to the left of ab. Hence obtain the current in the 10 Ω resistor. 	7	CO2	BT4

4	In the following circuit, maximum power is transferred to load resistance R_L, then R_L is -----Ω.	7	CO2	BT4
5	For the given circuit, determine the number of branches, meshes and nodes. Also, state whether it is a planar network	2	CO1	BT2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand basics electrical circuits with nodal and mesh analysis.
CO 2	Appreciate electrical network theorems.
CO 3	Apply Laplace Transform for steady state and transient analysis.
CO 4	Determine different network functions.
CO 5	Appreciate the frequency domain techniques.